QoE Characterization for Video-On-Demand Services in 4G WiMAX Networks

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Talk Outline

- Motivation
- Preliminary Experiments
- Survey
  - Protocol Overview (RTP)
  - QoE Metrics
- Simulation / Experiments with Video Traces
  - ns2, evalvid
- Future Work & Conclusions

Motivation

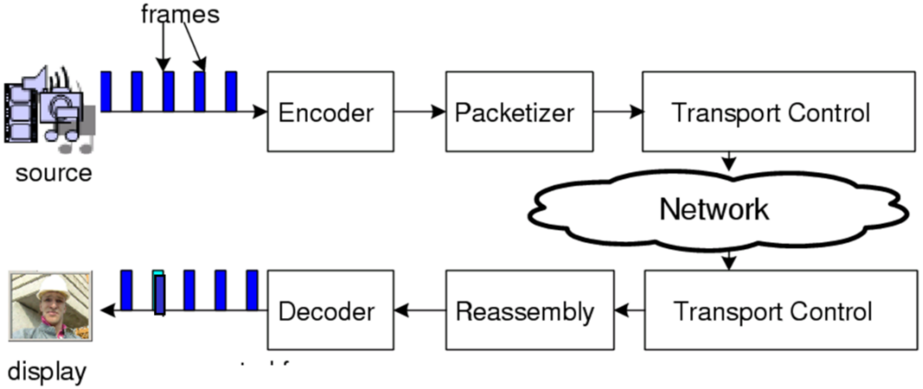
v Active deployment of triple and quadruple services

q ("The Fantastic Four": broadband internet access, television, and telephone with wireless service provisions).

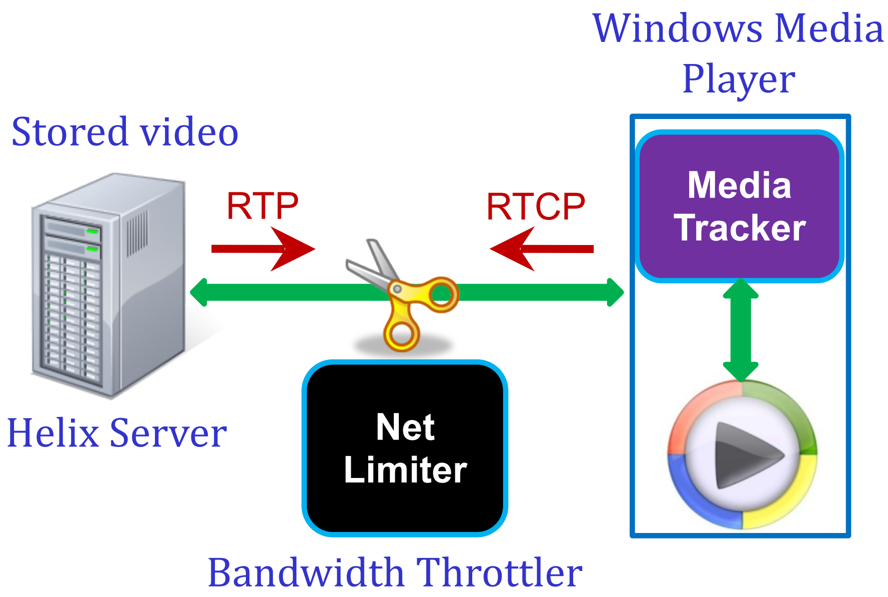
v Real-time, high-quality video and value-added data services over converged networks (e.g., 4G).

v Need to guarantee subscribers' Quality-of-Experience (QoE) and provide differentiated services in the face of heterogeneous end devices, varying wireless channels, resource constraints, etc.

v Lack of QoE orchestration models that map network events (e.g., variation in bandwidth, delay, packet error rates, jitter) to QoE.



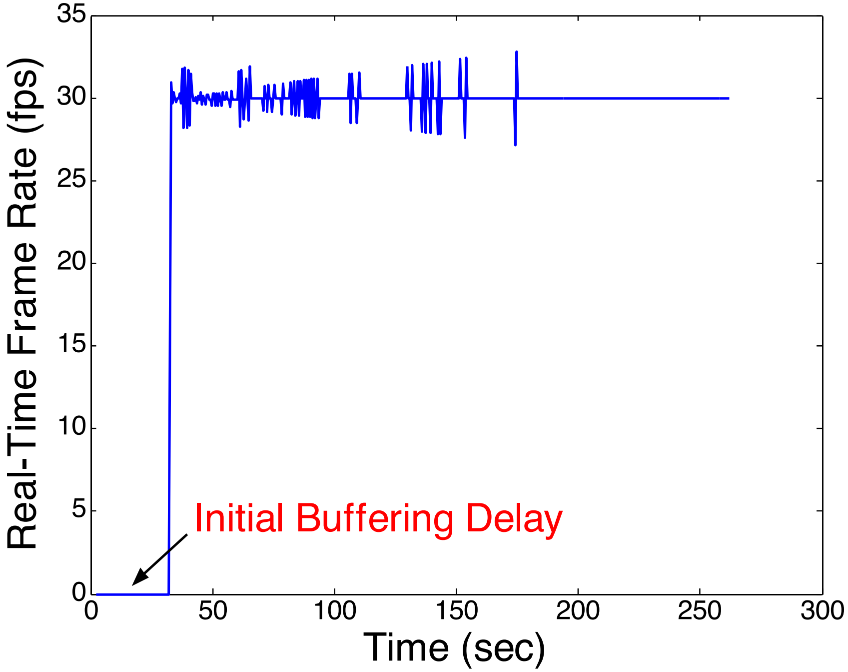
Preliminary Experiments



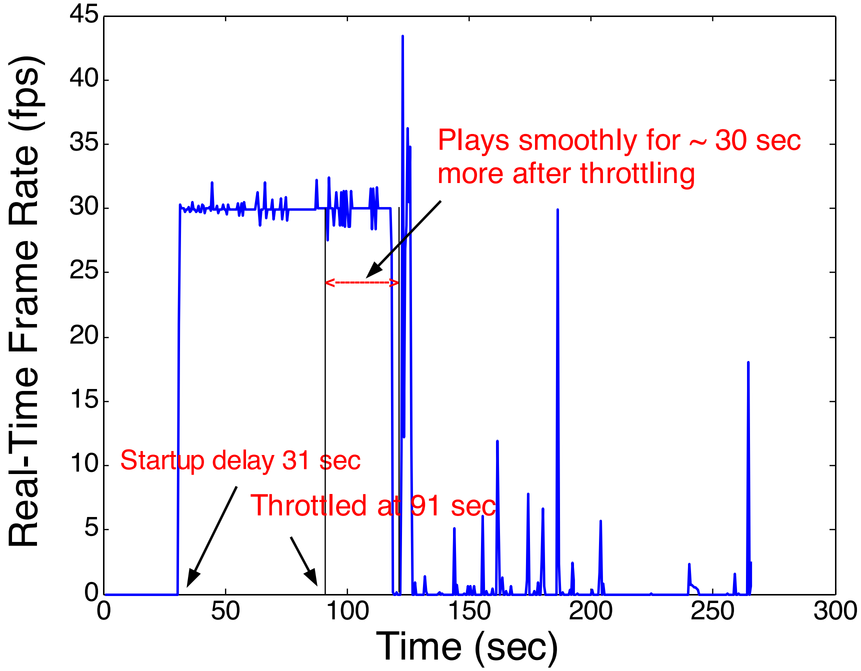
Video Characteristics

- Bit-rate: 1.96 Mbps
- Duration: 230 sec
- File size: 56.7 MB
- Media Player initial buffering: 30 sec

Preliminary Experiments



Without bandwidth throttling (no Net Limiter)

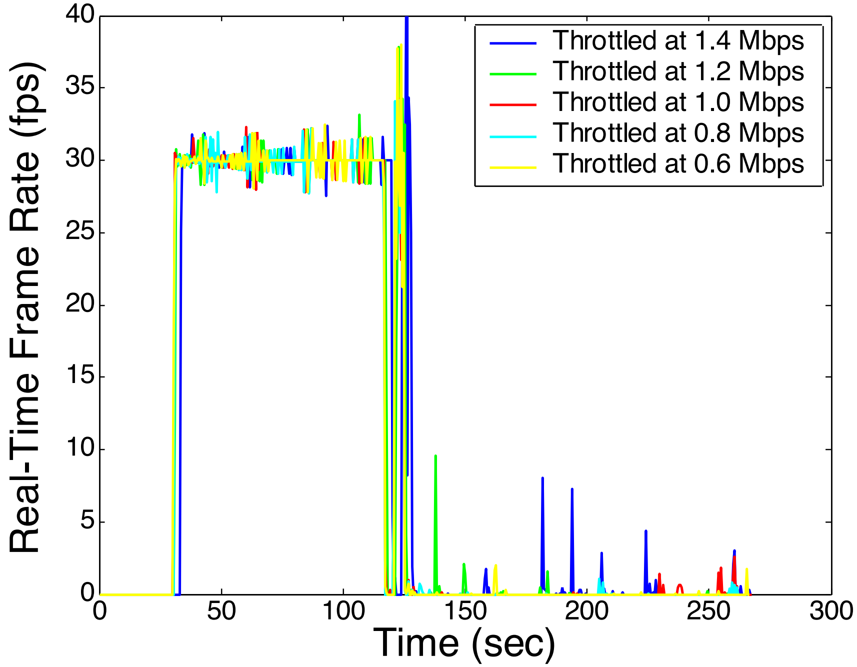


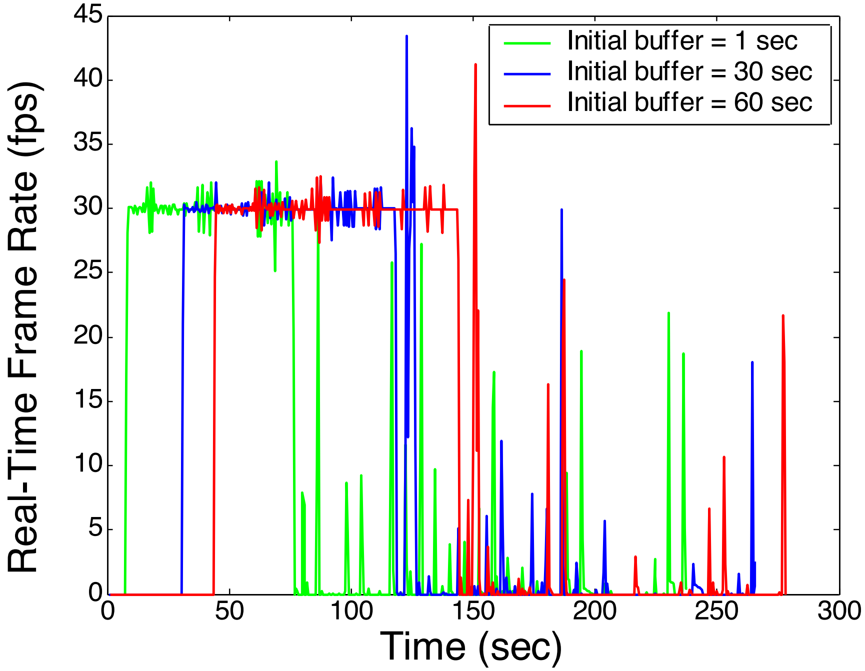
Bandwidth throttled to 1.6 Mbps at 60 sec from the start of the video

Observations

- Smooth video playback for initial buffering time after stalling
- Stalls frequently after 91 seconds, mimicking real-time frame rate

Preliminary Experiments





Variation of real-time frame rate with different initial buffering and throttling bandwidths

Observations

- Stalls frequently after 91 seconds irrespective of throttled bandwidth
- Small bandwidth variation and packet losses cause major degradation in QoE

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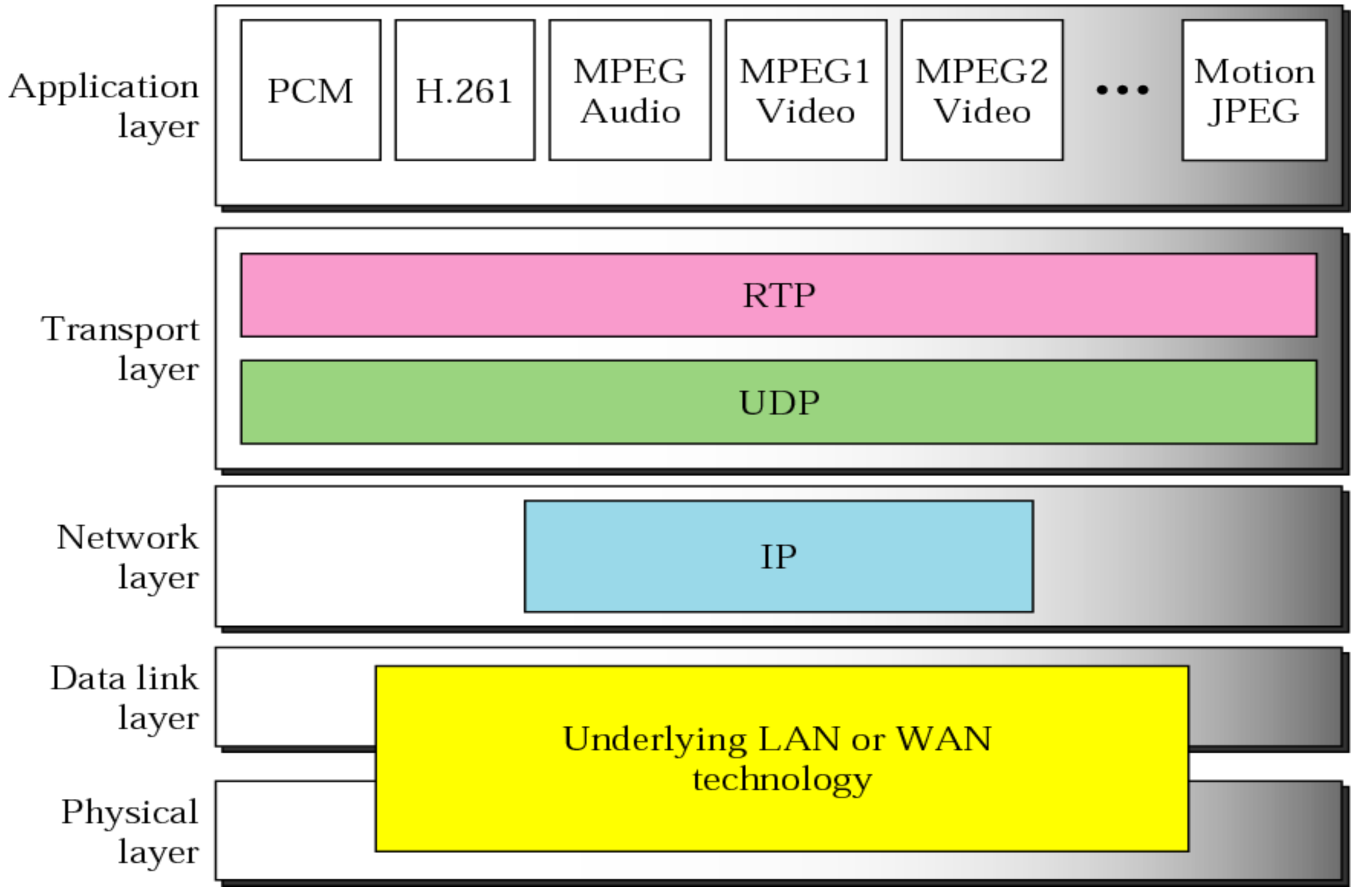
  - Protocol Overview (RTP, RTCP)
  - QoE Metrics
- Simulation / Experiments with Video Traces
  - ns2, evalvid, VLC, RTP dump
- Future Work
- Conclusions

Protocol Stack for Multimedia Services



Source:

RTP (Real-Time Transport Protocol)

- Provides end-to-end transport services for data with real-time characteristics
  - Payload type identification, sequence numbering, time stamping, delivery monitoring
- Does NOT provide timely delivery or other QoS guarantees
  - Relies on other protocols like RTCP and lower layers
- Does NOT assume the underlying network is reliable and does NOT deliver PDUs in sequence
  - Uses sequence number for reconstructing
- Application level framing
  - Headers can be modified and/or added to provide information required by applications
- Profile and Payload Format Specification Document
  - Defines a set of payload type codec and their mapping to payload formats
  - Defines how a particular payload is fragmented and mapped in RTP packets (RFC 3016 for MPEG-4)

RTP Header	VS Header	VO Header	VOL Header	Video Packet

RTP packet containing the configuration information and a video packet

RTP Header

Sampling instant of first data octet

- multiple PDUs can have same timestamp
- not necessarily monotonic
- used to synchronize different
- media streams

Payload type

Incremented by one for each RTP PDU:

- PDU loss detection
- Restore PDU sequence

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|V=2|P|X|  CC   |M|     PT      |       sequence number         |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                           timestamp                           |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|           synchronization source (SSRC) identifier            |
+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+
|            contributing source (CSRC) identifiers             |
|                             ....                              |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

- Each source of RTP PDUs; unique random 32-bit ID (SSRC)
- Packets with the same SSRC shares the same timing & sequence number space so a receiver groups packets by SSRC for playback

Contributing sources
(used by mixers)

Attributes of QoE

- Session Quality
  - Users' overall experience
    - especially from a connection perspective
  - Most affected by
    - initial buffering, re-buffering during playback, audio-video synchronization, packet losses, buffer over/under flow, codec, CPU limitation
- Video Quality
  - Frame quality, fidelity/smoothness of motion (fps), stalling
- Audio Quality
  - Fidelity and Mono/Stereo
    - Users may have different perceptions of what they are seeing based on what they are hearing
  - Different kinds of content need different levels of audio
- Nature of Content
  - Contributes to the weight of each factor in shaping QoE

    e.g., sports video may require video smoothness over picture clarity;
    a talking news head may require better audio than picture quality

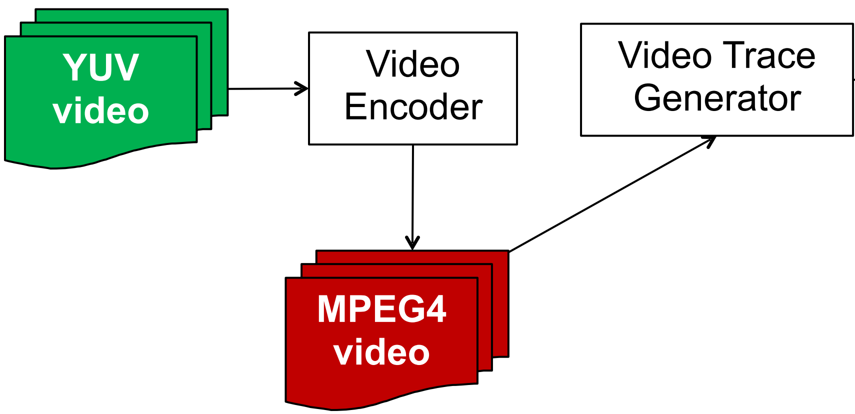
Video Evaluation Schemes & QoE Metrics

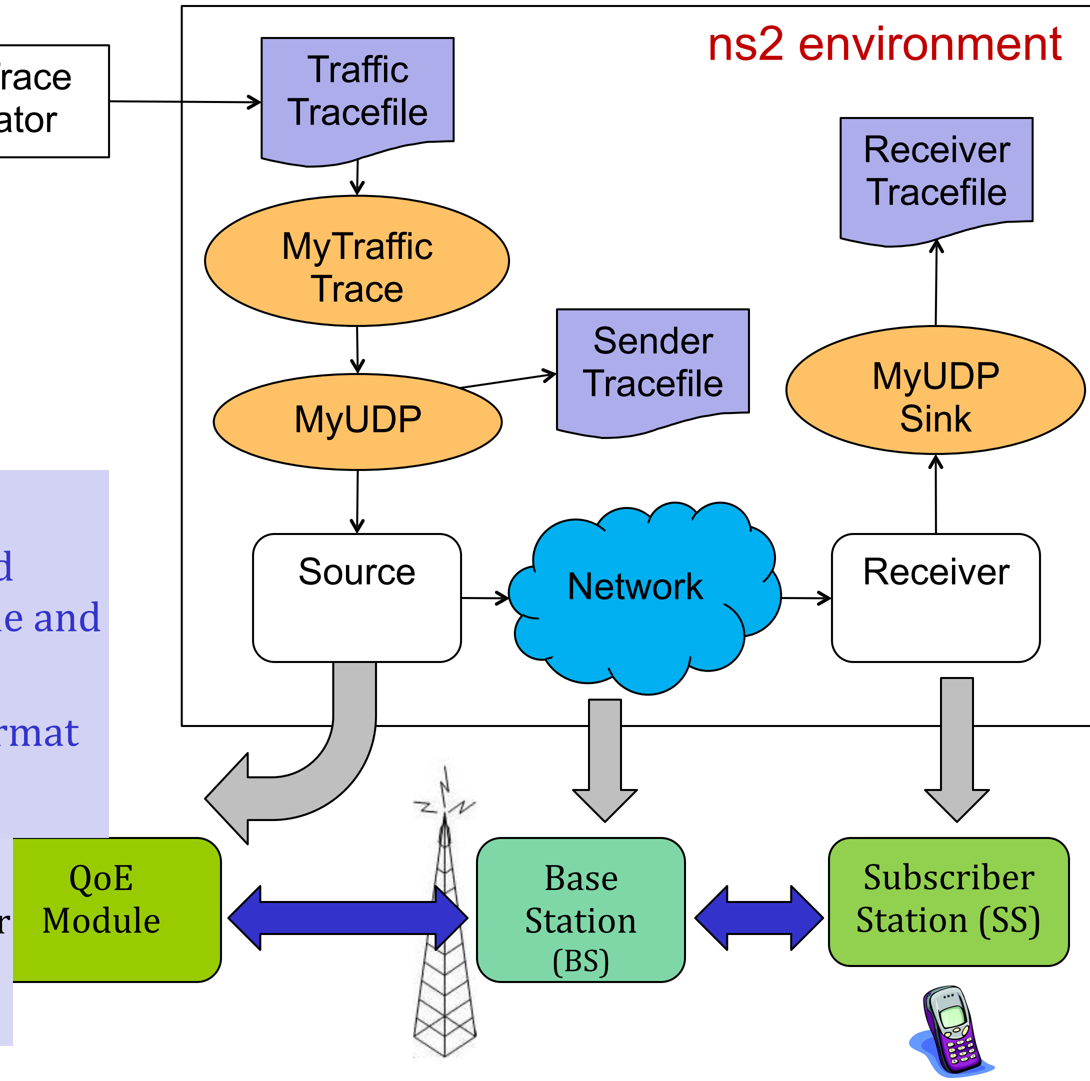
- End System Based
  - Primarily developed to evaluate various transcoding schemes
  - Characterize stream after network transmission is done
  - Cannot isolate network induced impairments, thus cannot recover; various QoE Metrics
- Objective Metrics – based on mathematical models
  - PSNR (Peak Signal to Noise Ratio) – most widely used frame to frame calculation; does not correlate very well with human perception; does not take delay, jitter into account
  - UQI (Universal Quality Index) – based on structural attributes of objects in the scene; separates comparison of structure, luminance, and contrast
  - SSIM (Structural Similarity Index) – based on Human Visual System; improvement over UQI; starting to replace PSNR
- Subjective Metrics – based on human perception
  - MOS (Mean Opinion Score); VQM (Video Quality Metric); PEVQ (Perceptual Evaluation of Video Quality)

Today's Solution and What is Lacking?

- Error Concealment
  - Intends to conceal the visual effects of packet loss by exploiting temporal or spatial correlation with adjacent data
  - Picture quality may reduce keeping the number of frames constant
- Frame Skipping
  - Does not decode a frame unless all packets are received
  - Picture quality remains same, but stalls occur
- Bit-rate Capping & Switching, TCP, Feedback-based encoding, etc…
  - Streaming at a bit rate matching the capabilities of handset &network
  - Switch streams between different encoded rates
- What we need?
  - In-network elements that can detect events (variation in channel condition, packet error rates, delay, jitter), infer about the experience (QoE Model) and take preventive action (QoE Orchestration) to maintain video quality

Simulation Setup: ns2 + Evalvid





Other Features of Evalvid

- Generate received video in compressed format (MPEG4) from receiver trace file and the original video
- Decode compressed video into YUV format
- Compute PSNR

segments

- Sends the segments to the lower UDP layer at appropriate times

Evaluation: Sony Video

Video Characteristics:

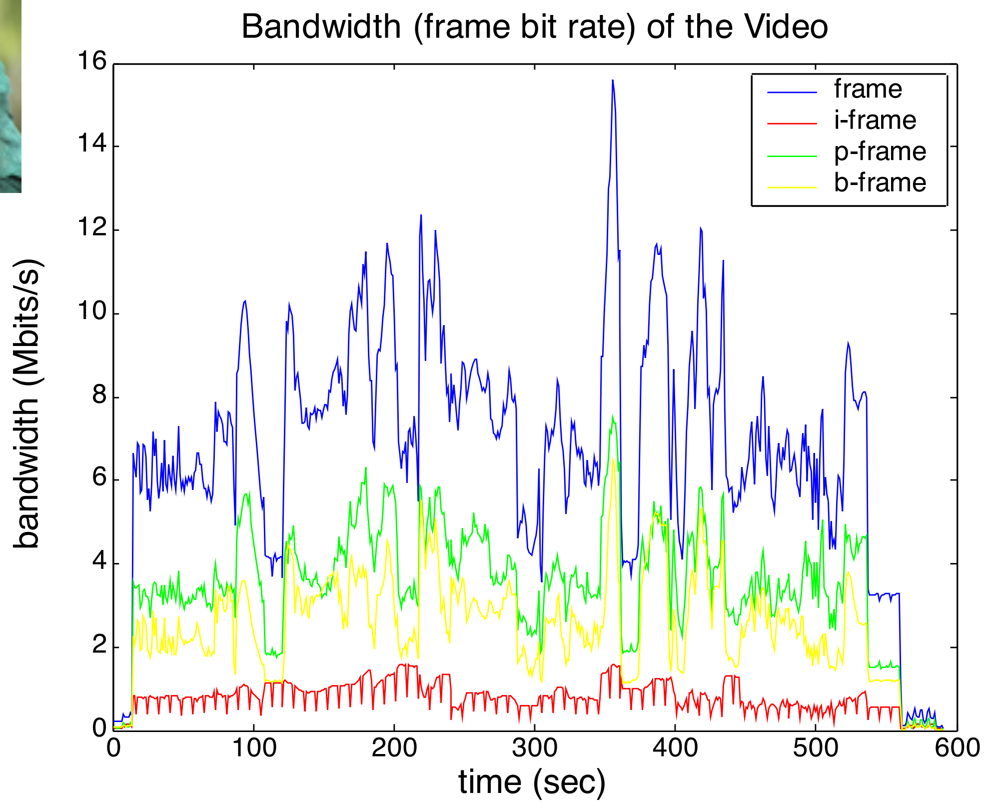
Encoder: MPEG-4
Variable Bit Rate (VBR)
Frame Size: CIF 352x288
No. Frames: 17681
GoP Size: 16
(IPPBPPBPPBPPBPPB)

Number of i-frames: 1106
Number of p-frames: 7735
Number of b-frames: 8840

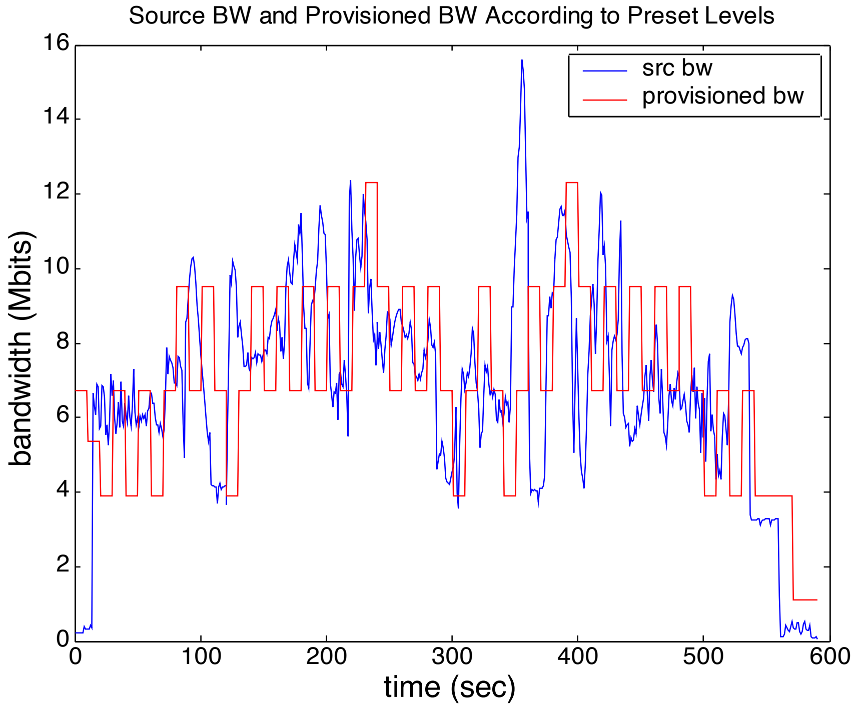
Video duration: 589 sec (~10 min)

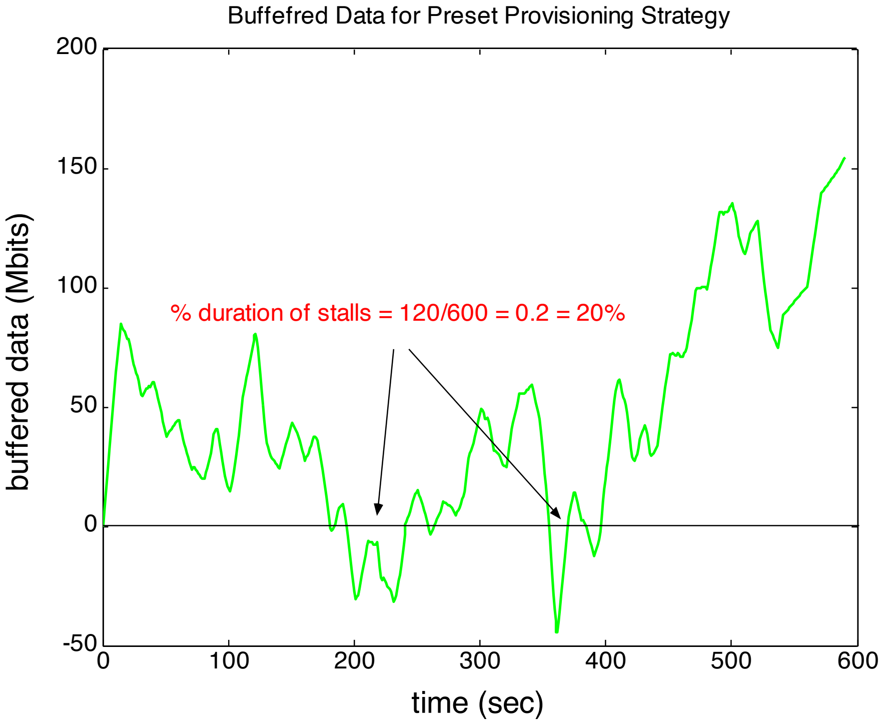
Mean frame size: 0.2256 Mbits
SD frame size: 0.1266 Mbits

Mean bandwidth: 6.727 Mbits/s
SD bandwidth: 2.8 Mbits/s



The I-frames occupy only ~20% of the mean bandwidth. One can assign higher priorities to the I-frames and let those packets pass when the wireless channel is bad.





Preset BW Provisioning:

Provisioned bandwidth levels are preset at the following values:

$(\mu-2\sigma) = 1.10$, $(\mu-\sigma) = 3.90$, $(\mu-0.5\sigma) = 5.36$, $\mu = 6.72$, $(\mu+0.5\sigma) = 8.12$, $(\mu+\sigma) = 9.52$, $(\mu+1.5\sigma) = 10.92$, $(\mu+2\sigma) = 12.32$, $(\mu+2.5\sigma) = 13.72$, $(\mu+3\sigma) = 15.12$ Mbits/s

What is needed : We need a dynamic provisioning of bandwidth, instead of preset levels, so we can withstand the variation in source BW and never let the buffer go empty or negative. Also the buffer content should be above a minimal threshold and should not store unnecessary data.

Strategy: Calculate the average source BW at every 10 sec window (configurable). If the calculated average BW is greater than the current provisioned BW, then provision a BW that is two levels higher than the current. Else provision a BW that is two levels lower than the current.

Evaluation: Silence of the Lambs Movie

Video Characteristics:

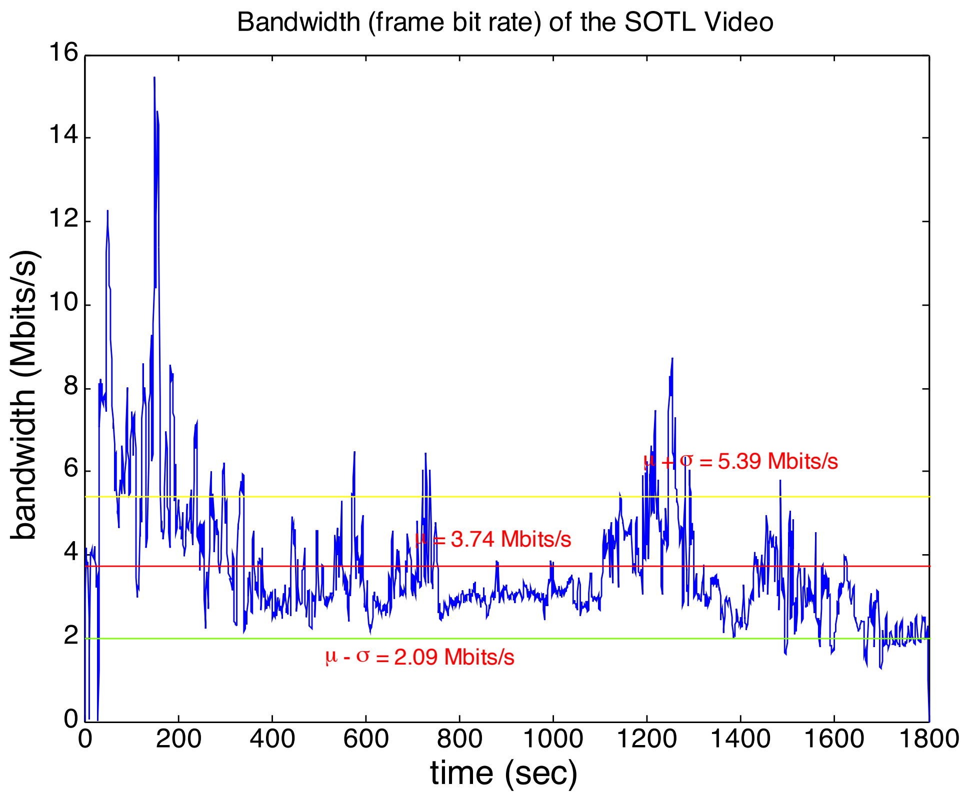
Encoder: MPEG-4
Variable Bit Rate (VBR)
Frame Size: CIF 352x288
No. Frames: 53953
GoP Size: 16
No. B Frames: 1

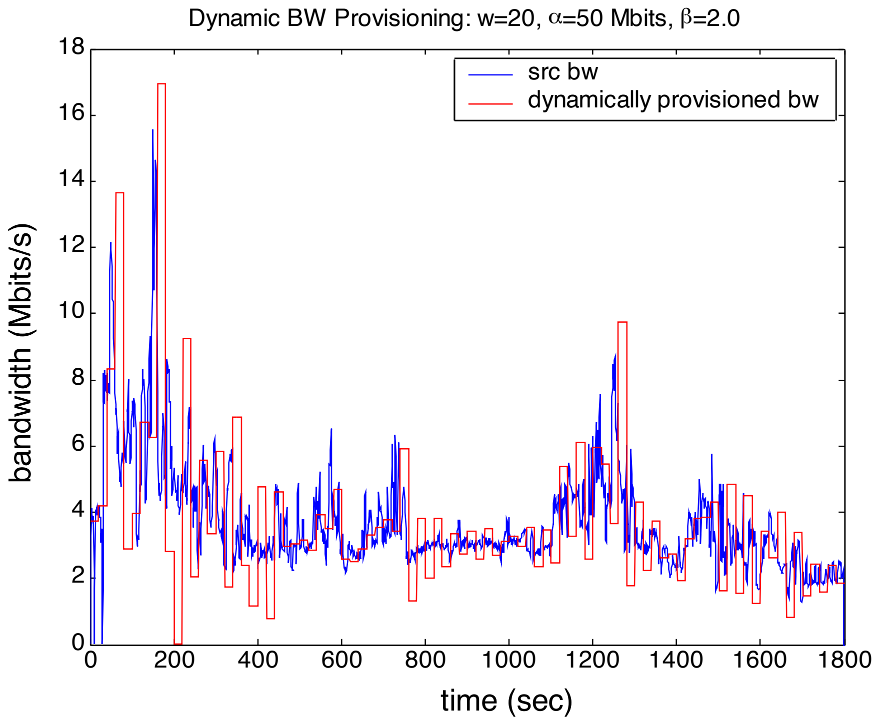
Number of i-frames: 3373
Number of p-frames: 23606
Number of b-frames: 26974

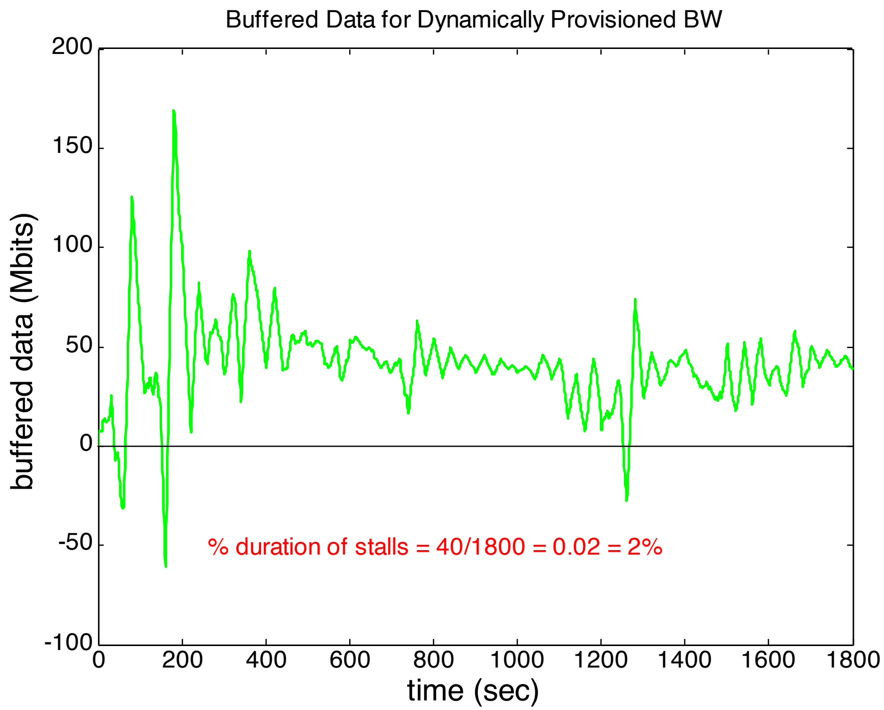
Video duration: 30 min (1800 sec)

Mean frame size: 0.1227 Mbits

Mean bandwidth: 3.74 Mbits/s
SD bandwidth: 1.65 Mbits/s







Dynamic Provisioning of BW

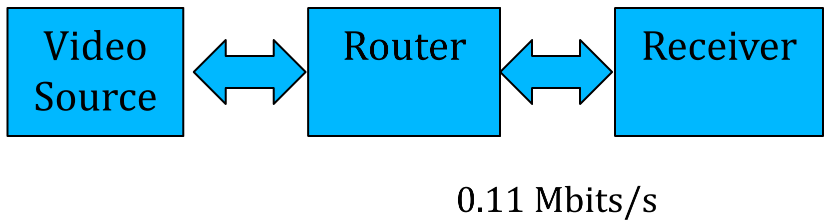
- Bandwidth levels are ***NOT*** preset as before
- Calculate the cumulative moving average $\mu(i)$ and standard deviation $\sigma(i)$ of source BW for a given window size and provision a BW according to the following:

$$BW(i+1) = \mu(i) + k.\sigma(i) + \alpha/w - \beta.\left[\sum_j w.BW(j) - w.i.\mu(i)\right]/w$$

Summary & Future Work

- Ground Work
  - Survey of VoD literature, protocols, and QoE metrics for VoD services
  - Developed simulation framework using ns2, evalvid, etc
- QoE Model
  - Identifies the time instants at which video gets stalled and correlate them with source BW
  - Predict QoE (no. of stalls, spread / distribution of stalls, duration of stalls, amount of contingency buffer, etc) for a given time and BW provisioning strategy
- QoE orchestration
  - First cut dynamic provisioning of BW based on tracking the source bit rate; this works better than preset level-based provisioning
  - There exists a trade-off between the number of stalls, the spread of stalls, the average duration of stalls, and the amount of buffer content with BW update frequency
    - Average duration of stalls increases as update frequency is decreased
    - Amount of buffered content is inversely proportional to the BW update frequency
- Improve upon the dynamic provisioning strategy
  - Use of Control Theory
- Multiple subscriber stations
- Incorporate
  - Wireless channel model
  - Constraints on BW provisioned between BS-SS

Demo



Video Characteristics

q 30 fps VBR
q Total # of frames: 401
q Video duration: 13 sec
q Mean bandwidth: 0.11 Mbits/s
q SD bandwidth: 0.02 Mbits/s